

LAB 5: ENTERPRISE WINDOWS SERVER ADMINISTRATION

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Group 3

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EXECUTIVE SUMMARY

WoodRock proved continued company growth by having prioritized specific goals to reduce time spent on internal tasks and offer increased efficiency in employee management. The company sought to prioritize company executive print jobs on the sole network printer, which was completed by the addition of another print queue. The original queue was configured for only administrators at the highest priority level, while the new queue had a duplex unit installed for double-sided print jobs. WoodRock promoted an employee to a print operator for the sake of management.

With WoodRock's growth, the company could not be limited by distance and accessibility with configuration. WoodRock enabled PowerShell remoting on server virtual machines for remote operating system services management. The addition gave WoodRock the ability to start and stop services from a remote operating system. To increase remote capabilities, WoodRock implemented Windows Admin Center (WAC) for the remote management of all Windows machines across the Virtual Machines (VMs) from G03SRV06. The admin center was available to domain and non-domain devices from the browser.

WoodRock sought a major improvement to data-access efficiency, and Distributed File System (DFS) was a network accessible way to hold/share company data. The network share provided WoodRock employees an easy way to utilize operational data held within two replicated namespace folders.

Lastly, WoodRock needed an efficient and automated way to preserve the WoodRock server environment. Windows Server Update Services (WSUS) would be implemented on G03SRV07 to schedule patches and updates across all Windows VMs, which greatly reduced the effort required to manually apply all server patches.

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BUSINESS SCENARIO

The wood sales company WoodRock was looking to increase sales and ease management across the enterprise. WoodRock had incorporated virtualization using two Nutanix Acropolis Hypervisors (AHV) embedded on two separate Dell Pro Tower Plus machines. Each hypervisor hosted WoodRock's own cluster holding a variety of virtual machines, where WoodRock employees were able to operate. The clusters G03NUTXSRV01 and G03NUTXSRV02 were home to Windows Server 2025 and Windows 11 images booted on the VMs, and one VM was booted with a TrueNAS image. The physical architecture of WoodRock's system offered an environment for employees to operate at a much lower infrastructure cost, which gave WoodRock administrators centralized access to each operating system (OS) through Prism Central.

WoodRock needed to address a problem with print job traffic on the k238 (HP LaserJet E40040) printer queue on G03SRV01VM. The first order of action was to add a print operator, Hannah Doll, to re-order, delete, and edit print jobs. WoodRock additionally added another printer in the print management tool, which offered a second printer queue to have pointed to the same physical printer at 44.3.1.238. The new print queue called k238-duplex held a duplex unit for double-sided print jobs, where the original print queue was configured to have higher priority for specifically administrators. Overall, WoodRock addressed common traffic issues over the print server to have provided efficiency in printed documents.

With services having been available remotely through PowerShell, WoodRock's interests aligned with the ability to list, stop, and start services from a remote OS. G03SRV03VM enabled PowerShell remote session connections, with G03SRV01VM having hosted the connection to

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G03SRV03VM. Ease of service management allowed WoodRock employees to operate from distant operating systems.

WoodRock had a need for network accessible files across the enterprise, with the best way to have done so having implemented a distributed file system in the forest. DFS was installed on G03SRV01VM, with the namespace server \\group03.c24200.cit.lcl\dfs and two namespaces called RedirectedFolders and RoamingProfiles. The replication was across G03SRV05 with the replicated folders being both RedirectedFolders and RoamingProfiles. WoodRock had successfully incorporated network accessible data for efficient access by employees, where the folders in the namespace were targets to external company data.

WoodRock desired for efficiency among all Windows operating systems across company VMs. A new VM G03SRV07 was implemented for the purpose of having configured Windows Server Update Services, where the server was set to automatically update all tools for all computer VMs. In addition, Windows Admin Center was configured and installed on G03SRV06 to allow remote management of all Windows VMs across the enterprise, where accessibility was ensured by the domain environment at <https://G03SRV06> and by the external environment at <https://44.38.3.55>. WoodRock's quarterly revenue had increased as a result of efficient internal operations and ease of access across network accessible data.

Figure 1: Logical Model

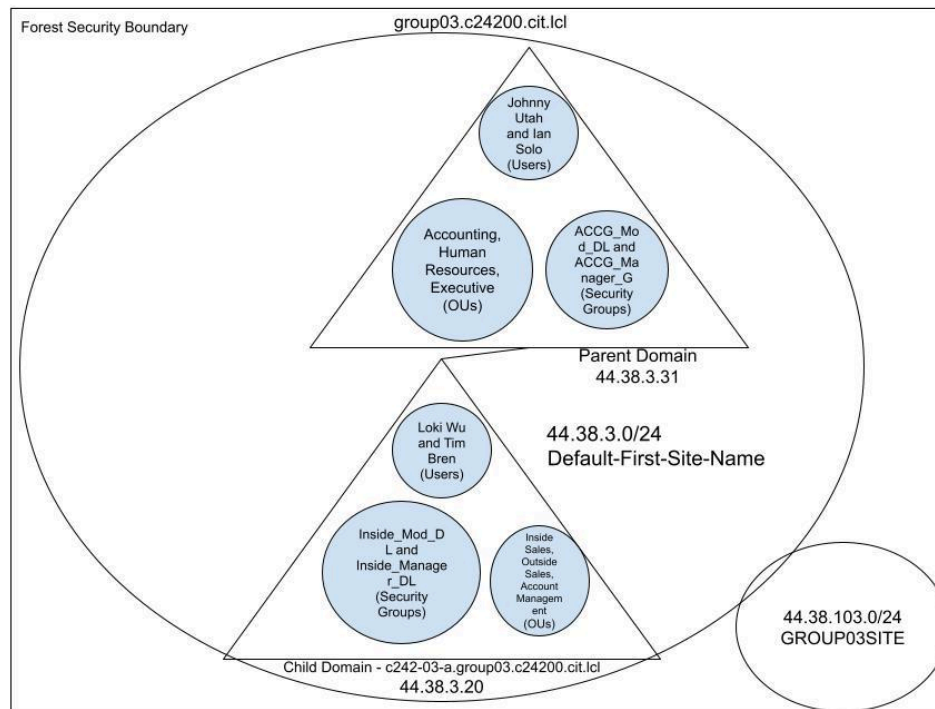


Figure 1.) Logical model of WoodRock's forest pre-execution of procedures

The WoodRock forest was emulated with the logical model of both the parent and child domains. The parent domain was home to three organizational units which were Accounting, Human Resources, and Executive for the respective WoodRock departments. As for the child domain, Inside Sales, Outside Sales, and Account Management were the organizational units (OUs) hosted for the given departments. Users Johnny Utah and Ian Solo were exceptional employees of the Accounting department, and were given entry to the ACCG_Mod_DL security group. In the child domain, users Loki Wu and Tim Bren had proven excellence and experience which gave entry to the Inside_Manager_DL security group. Replication was apparent between both sites in the parent domain, with GROUP03SITE not having hosted a domain.

Figure 2: Physical Model

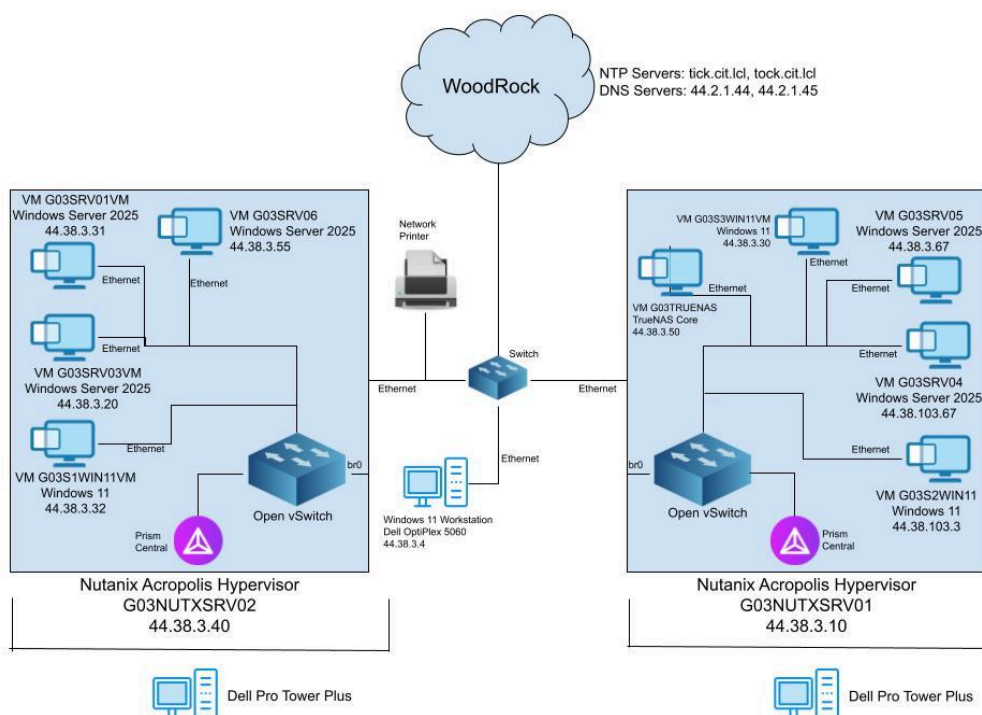


Figure 2.) Physical model of WoodRock topology pre-execution of procedures

WoodRock's environment topology continued to expand with five virtual machines hosted on the G03NUTXSRV01 cluster and four virtual machines on the G03NUTXSRV02 cluster. The two Nutanix AHVs were held on two separate Dell Pro Tower Plus machines, where the virtual machines held within were connected via an Open vSwitch to the network printer, Dell OptiPlex 5060, and the WoodRock computer information technology (CIT) network. All connections in the topology were via ethernet. The network provided Domain Naming Servers (DNS) and Network Time Protocol (NTP) servers for VM reference.

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PROCEDURES

From a technical point of view, WoodRock took steps for a scalable and centrally managed network environment to be built. First, the organization established a robust data management strategy with the installation and configuration of the Distributed File System alongside Group Policy Objects (GPOs) for automated folder redirection and roaming profiles. Next, the administrators deployed a comprehensive printing infrastructure on the primary Domain Controller (DC), which included dedicated administrative accounts creation, network-capable hardware installation, and specialized duplex print queues setup. As the business infrastructure grew, individual VMs management became more complex, which led to the enablement of PowerShell Remoting across the entire fleet for streamlined command-line administration. Furthermore, the environment was modernized with the installation of WAC which provided a unified web-based management interface. Additionally, all Windows Server 2025 and Windows 11 machines were integrated into the dashboard through the previously established Windows Remote Management (WinRM) exceptions. Finally, the configurations were verified and enforced through Active Directory (AD) domain-wide policy updates, which ensured consistent performance across the Nutanix-hosted infrastructure.

Each section contained a numbered list which showed the steps. Sections' headers, which were bolded and centered, showed the actions taken. The bibliography appeared at the end of the report, followed by Appendix A, which contained the problem statement. Additional supporting documentation was included in subsequent appendices.

The followings were the formatting keys: **buttons** were bold, *options* were italicized, text entered into the computer was in Courier New, menu navigation was by the pipe symbol and italic words: *Start | Programs | MS Office | Word*.

G03SRV07 Deployment on Prism Central

The section detailed the deployment of G03SRV07 from the G03WIN2025TEMP template. Deployment from a template provided a standardized configuration for the VMs in the environment and reduced setup time. The G03SRV07 deployment was a critical component of business continuity and security, as a host server for Windows Server Update Services. As a host server for WSUS, the deployment of G03SRV07 served as a cornerstone for WoodRock's operational security, which regulated patch distribution and safeguard against the widespread downtime caused by unverified updates. The 500-GigaByte (GB) storage disk also provided enough capacity to maintain a comprehensive update repository for all domain assets. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Compute* dropdown | *VMs* | **Create VM from Template** | *Please Select* dropdown | *G03WIN2025TEMP* | **Begin**
2. Typed G03SRV07 in *Name* field
3. Selected *Cluster* dropdown | *G03NUTXSRV01* | **Advanced Deploy** | **Next** in *Configuration* window
4. Deselected *G03NUTXSRV02* in *Networks* window
5. Clicked **Attach to Subnet** | **Save** in *Attach to Subnet* window | **Next** in *Resources* window | **Next** in *Management* window | **Deploy** in *Review* window
6. Pressed **Ctrl + R**
7. Clicked *G03SRV07* | **Update** | **Next** in *Configuration* window | *Attach Disk* | *Storage Container* dropdown | *G03VM*
8. Typed 500 in *Capacity* field

9. Clicked **Save** in *Attach Disk* window | **Save** in *Deploy VM from Template* window | **More**
| *Power On*
10. Pressed **Ctrl + R**

Logging into G03SRV07

The section detailed how the VM console from the Nutanix Compute dashboard was started and logged in where the administrative credentials were used. Consistent access to the VM console was essential for work related to G03SRV07 and a fundamental requirement for the ongoing administration of the WSUS server. For WoodRock, access was critical for low-level system maintenance, network connectivity troubleshooting, and the repository update management without relying on network-dependent protocols like Remote Desktop Protocol (RDP). The following steps were documented on G03SRV03.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge*
Searchbar field
3. Pressed **Enter**
4. Clicked *Compute* dropdown | *VMs* | *G03SRV03VM* | *Launch Console* | **Send CtrlAltDel**
5. Typed `password` in *Password* field
6. Pressed **Enter**

Internet Protocol Address Configuration on G03SRV07

The section detailed the process of Internet Protocol (IP) version 6 (IPv6) disabling and static IP version 4 (IPv4) address (44.38.2.20) and DNS configurations for G03SRV07. Proper IP configuration allowed the new server to be communicated with WoodRock's DCs and services. For WoodRock, the configuration was a mechanical necessity. As G03SRV07 server was

destined to host WSUS, any change in the IP address prevented security patches being pulled from every workstation and server in the enterprise. As a static IP address was assigned, WoodRock ensured the configuration centralized update repository remained a reliable "known" target, which prevented the internal network congestion and security vulnerabilities when update services became unreachable. **Refer to Appendix B: System and Network Configuration.**

Refer to Appendix B: System and Network Configuration.

1. Clicked *Server Manager* in *Windows Taskbar* | *Local Server* | *IPv4 address assigned by DHCP, IPv6 enabled*
2. Double-selected *Ethernet*
3. Clicked **Properties**
4. Deselected *Internet Protocol Version 6 (TCP/IPv6)*
5. Double-clicked *Internet Protocol Version 4 (TCP/IPv4)*
6. Clicked *Use the following IP addresses* | *Use the following DNS server addresses*
7. Typed 44.38.3.70 in *IP address* field
8. Inputted 255.255.255.0 in *Subnet mask* field
9. Typed 44.38.3.31 in *Preferred DNS server* field
10. Inputted 44.38.3.1 in *Default gateway* field
11. Clicked **OK** in *Internet Protocol Version 4 (TCP/IPv4) Properties* window | **OK** in *Ethernet Properties* | **Close** in *Network Connections* window

Computer Name and Domain Configuration on G03SRV07

The section detailed how G03SRV07 was enrolled from a workgroup to the group03.c24200.cit.lcl domain. The process allowed the server to participate in domain-based security, Group Policy, and centralized management. For WoodRock, domain enrollment, which

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provided the centralized security and management necessity, ensured consistent user access and organizational control over all network assets. **Refer to Appendix B: System and Network**

Configuration.

1. Clicked *Server Manager* in *Windows Taskbar* | *Local Server* | *G03WIN2025TEMP* in *Computer name* window | **Change...** | *Domain*
2. Typed G03SRV07 in *Computer name* field
3. Inputted group03.c24200.cit.lcl in *Domain* field
4. Clicked **OK** in *Computer Name/Domain Changes* window
5. Typed Administrator in *User name* field
6. Inputted password in *Password* field
7. Clicked **OK** in *Windows Security* window | **OK** in *Computer Name/Domain Changes* window | **Restart Now** in *Microsoft* window

WSUS Folder Creation on G03SRV07

The section detailed how the second 500 GB virtual disk was configured and formatted with the default file system, New Technology File System (NTFS), where updates resided. For WoodRock, the update repository was kept on a separate volume, which was a critical best practice and prevented the primary system drive capacity from being exceeded as the library of Microsoft patches grew over time. WoodRock ensured the server remained operational even during high-volume synchronization tasks, which maintained the stability of the WSUS service and prevented storage-related failures. **Refer to Appendix B: System and Network**

Configuration.

1. Right-clicked **Start**
2. Selected *Disk Management*

3. Right-clicked *Disk 1*
4. Selected *Online*
5. Right-clicked *Disk 1*
6. Selected *Initialize Disk* | **OK** in *Initialize Disk* window
7. Right-clicked *Unallocated space*
8. Selected *New Simple Volume...* | **Next** in *Welcome to the New Simple Volume Wizard* window | **Next** in *Specify Volume Size* window | **Next** in *Assign Drive Letter or Path* window
9. Typed *WSUS* in *Volume label* field
10. Clicked **Next** in *Format Partition* window | **Finish** | **Close** in *Disk Management* window
11. Selected *File Explorer* in *Windows Taskbar* | *WSUS (F:)* | *New Folder*
12. Typed *WSUS* in *folder's name* field
13. Pressed **Enter**

WSUS Installation and Configuration on G03SRV07

The section detailed the installation of the WSUS role and the subsequent configuration of the update synchronization pipeline. For WoodRock, the procedure established the administrative engine required for approval and distribution of Microsoft patches across the AD domain. As the content location was directed to the dedicated F:\WSUS directory and the “WsusUtil” Secure Socket Layer (SSL) configuration was executed, WoodRock ensured update traffic was both locally cached and securely transmitted. The centralized control was vital for a hardened security posture maintenance, which made a uniform update schedule enforcement possible while the network bottlenecks were prevented from unmanaged, individual machine downloads. **Refer to Appendix B: System and Network Configuration.**

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1. Clicked *Server Manager* in *Windows Taskbar* | *Local Server* | *Manage* | *Add Roles and Features* | **Next** in *Before You Begin* window | **Next** in *Installation Type* window | **Next** in *Server Selection* window
2. Selected *Windows Server Update Service* | **Add Features** | **Next** in *Server Roles* window | **Next** in *Features* window | **Next** in *Role Services* window
3. Typed `F:\WSUS` in *Content location selection* field
4. Clicked **Next** in *Content* window | **Next** in *Web Server Role (IIS)* window | **Next** in *Role Services* window | **Install** | **Close** in *Results* window | **Notifications** in *Server Manager* window | *Launch Post-Installation tasks*
5. Selected *Tools* in *Server Manager* window | *Windows Server Update Services* | **Next** in *Before You Begin* window | **Next** in *Microsoft Update Improvement Program* window | **Next** in *Choose Upstream Server* window | **Next** in *Specify Proxy Server* window
6. Clicked **Start Connecting** | **Next** in *Connect to Upstream Server* window | **Next** in *Choose Languages* window | **Next** in *Choose Products* window | **Next** in *Choose Classifications* window | **Next** in *Configure Sync Schedule* window
7. Selected *Begin initial synchronization* | **Next** in *Finished* window | **Finish** | **Close** in *Update Services* window
8. Clicked *Command Prompt* in *Windows Taskbar*
9. Typed `cd "C:\Program Files\Update Services\Tools"` in `C:\Users\TEMP>` field
10. Pressed **Enter**
11. Typed `C:\Program Files\Update Services\Tools>WsusUtil.exe configuressl G03SRV07` in `C:\Program Files\Update Services\Tools>` field

12. Pressed **Enter**

Computer Group Creation for WSUS on G03SRV07

The section detailed how the WSUS Computers logical group was created in the update management console. At WoodRock, grouping network assets was an important step in granular patch management where updates were distributed by administrators to specific sets of machines rather than a blanket policy being applied to every server and workstation at the same time. The organizational structure was important for business continuity where patches were allowed to be rolled out in phases and updates to be tested on a subset of the infrastructure before a full-scale deployment. As WSUS Computers group was created, the business created the necessary framework to prevent widespread system failures caused by untested or incompatible software updates. **Refer to Appendix B: System and Network Configuration.**

1. Selected *Server Manager* in *Windows Taskbar* | *Tools* | *Windows Server Update Services* | *Update Services* | *G03SRV07* | *Computers*
2. Right-clicked *All Computers*
3. Clicked *Add Computer Group...*
4. Typed `WSUS Computers` in *Name* field
5. Clicked **Add** in *Add Computer Group* window | **Close** in *Update Services* window

WSUS GPO Setup on G03SRV01VM

The section detailed how the GPO was set up on the DC for update automation within the enterprise. For WoodRock, the GPO guaranteed the use of the new WSUS server for WoodRock and prevented any assets joined to the domain from outside patches lookup of Microsoft servers. As updates location and automatic schedule installation was centrally defined through the Default Domain Policy, the business removed the risk of non-compliant individual workstations

due to user error. The automated governance was critical for a hardened security and ensured critical patches were applied consistently across the infrastructure without manual intervention.

Refer to Appendix B: System and Network Configuration.

1. Selected *Server Manager* in *Windows Taskbar* | *Tools* | *Group Policy Management*
2. Right-clicked *Default Domain Policy*
3. Selected *Edit...* | *Computer Configuration* | *Policies* | *Administrative Templates: Policy definition* | *Windows Components* | *Windows Update* | *Manage end user experience*
4. Double-clicked *Configure Automatic Updates*
5. Selected *Enabled* | *Configure automatic updating* dropdown | *Auto download and schedule the install* | **Apply** | **OK** in *Configure Automatic Updates* window | *Manage updates offered from Windows Server Update Service*
6. Double-clicked *Specify intranet Microsoft update service location*
7. Selected *Enabled*
8. Typed `http://G03SRV07:8530` in *Set the intranet update service for detecting updates* field
9. Inputted `http://G03SRV07:8530` in *Set the intranet statistics server* field
10. Clicked **Apply** | **OK** in *Specify intranet Microsoft update service location* window | **Close** in *Group Policy Management Editor* window | *Powershell* in *Windows Taskbar*
11. Typed `gpupdate /force` in *C:\WINDOWS\system32* field
12. Pressed **Enter**
13. Typed `Y` in *OK to log off? (Y/N)* field
14. Pressed **Enter**

DFS Installation on G03SRV01VM

The section detailed how the DFS was installed and set up on the G03SRV01VM DC. For WoodRock, DFS was a mechanical necessity for a simplified, unified entry point of all corporate file shares. As the physical location of server data, \\group03.c24200.cit.lcl\DFS, was packed into a single logical path, WoodRock ensured employees had access to critical resources without needing to know specific server names. The architecture was vital for business continuity where the framework was provided for future replication and high availability, which ensured accessibility of shared folders even if individual backend servers underwent maintenance or experienced hardware failure. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Windows Taskbar* | *Manage* | *Add Roles and Features* | **Next** in *Before You Begin* window | **Next** in *Installation Type* window | **Next** in *Server Selection* window
2. Selected *DFS Namespaces* | *DFS Replication*
3. Clicked **Next** in *Server Roles* window | **Next** in *Features* window | **Install** in *Confirmation* window | **Close** in *Add Roles and Features* window
4. Selected *Tools* in *Server Manager* window | *DFS Management* | *New Namespace* in *DFS Management* window | **Browse** in *Namespace Server* window
5. Typed G03SRV01VM in *Enter the object name to select* field
6. Clicked **Check Names** in *Select Computer* window | **OK** in *Select Computer* window | **Next** in *Namespace Server* window
7. Typed dfs in *Name* field
8. Clicked **Edit Settings** in *Namespace Name and Settings* window

9. Selected *Administrators have full access; other users have read and write permissions*
10. Clicked **OK** in *Edit Settings* window | **Next** in *Namespace Name and Settings* window | **Next** in *Namespace Type* window | **Create** in *Review Settings and Create Namespace* window | **Close** in *Confirmation* window

DFS Installation on G03SRV04

The section detailed how the DFS roles and services were installed on G03SRV04. The process expanded the organization of file management capabilities to another member server. For WoodRock, the DFS roles and services installation on G03SRV04 was a mechanical necessity for the storage infrastructure to be scaled where the G03SRV04 server participated in the enterprise-wide replication. As both DFS Namespaces and DFS Replication were installed, WoodRock prepared the G03SRV04 server to host redundant copies of critical data, which made sure WoodRock employees have redundant access to the shared storage even if another file server went down. The installation was an important part of the company's promise for the network to be highly available and for the single points of failure to be gotten rid of. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Windows Taskbar*
2. Selected *Manage* in *Server Manager* window | *Add Roles and Features* | **Next** in *Before You Begin* window | **Next** in *Installation Type* window | **Next** in *Server Selection* window
3. Clicked *DFS Namespaces* | *DFS Replication*
4. Selected **Next** in *Server Roles* window | **Next** in *Features* window | **Install** in *Confirmation* window
5. Clicked **Close** in *Add Roles and Features* window

New Shared Folders Creation inside DFS Namespace on G03SRV01VM

The section detailed how the actual folder structure was created in the DFS folder which held the data for WoodRock's users. The RedirectedFolders and RoamingProfiles folders creation was a key step for WoodRock for user environments centralization and data portability. As the folders were placed in the managed DFS path, WoodRock ensured employee documents and desktop settings were saved on the server and not on individual workstations, which were important for business continuity and disaster recovery. In the event of a physical workstation failure, any machine on the WoodRock network could be logged in by an employee for files and desktop settings immediate access, which ensured productivity without the risk of data loss on local computers. **Refer to Appendix B: System and Network Configuration.**

1. Selected *File Explorer* in *Windows Taskbar* window
2. Clicked *This PC | Local Disk (C:) | New folder |*
3. Typed *DFS*
4. Clicked *DFS | New dropdown | Folder*
5. Typed *RedirectedFolders* in *folder's name* field
6. Clicked *New dropdown | Folder*
7. Typed *RoamingProfiles* in *folder's name* field

New Shared Folders Creation inside DFS Namespace on G03SRV04

The section detailed how the actual folder structure was created in the DFS folder which held the data for WoodRock's users. The RedirectedFolders and RoamingProfiles folders creation was a key step for WoodRock for user environments centralization and data portability. As the folders were placed in the managed DFS path, WoodRock ensured employee documents and desktop settings were saved on the server and not on individual workstations, which were

important for business continuity and disaster recovery. In the event of a physical workstation failure, any machine on the WoodRock network could be logged in by an employee for files and desktop settings immediate access, which ensured productivity without the risk of data loss on local computers. **Refer to Appendix B: System and Network Configuration.**

1. Selected *File Explorer* in *Windows Taskbar* window
2. Clicked *This PC* | *Local Disk (C:)* | *New folder* |
3. Typed *DFS*
4. Clicked *DFS* | *New* dropdown | *Folder*
5. Typed *RedirectedFolders* in *folder's name* field
6. Clicked *New* dropdown | *Folder*
7. Typed *RoamingProfiles* in *folder's name* field

Share Permissions Configuration for Newly Created Folders on G03SRV01VM

The section detailed how the network share permissions were set for the enterprise data directories. For WoodRock, the use of *RoamingProfiles* and *RedirectedFolders* was an important security feature which ensured sensitive directories only accessible for the correct users as the network was browsed by employees. Full Control at the share level with the use of NTFS's granular permissions allowed the automated file transfers and user synchronization to be automated in the DFS environment. The mechanical requirement for a seamless user experience allowed data across the infrastructure to be saved and retrieved by WoodRock employees, which maintained the cleanliness and security of the network environment. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *File Explorer* in *Windows Taskbar* window | *This PC* | *Local Disk (C:)* | *DFS*
2. Right-clicked *RoamingProfiles*

3. Clicked *Properties* | *Sharing* | **Advanced Sharing** in *Sharing* window | *Share this folder*
4. Typed *RoamingProfiles\$* in *Share Name* field
5. Clicked **Permissions** in *Advanced Sharing* | *Everyone*
6. Selected *Allow Full Control*
7. Clicked **OK** in *Share Permissions* window | **OK** in *Advanced Sharing* window | **Close** in *RoamingProfiles Properties* window
8. Right-clicked *RedirectedFolders*
9. Clicked *Properties* | *Sharing* | **Advanced Sharing** in *Sharing* window | *Share this folder*
10. Typed *RedirectedFolders\$* in *Share Name* field
11. Clicked **Permissions** in *Advanced Sharing* window | *Everyone*
12. Selected *Allow Full Control*
13. Clicked **OK** in *Share Permissions* window | **OK** in *Advanced Sharing* window | **Close** in *RoamingProfiles Properties* window

Share Permissions Configuration for Newly Created Folders on G03SRV04

The section detailed how the network share permissions were set for the enterprise data directories. For WoodRock, the use of *RoamingProfiles* and *RedirectedFolders* was an important security feature which ensured sensitive directories only accessible for the correct users as the network was browsed by employees. Full Control at the share level with the use of NTFS's granular permissions allowed the automated file transfers and user synchronization to be automated in the DFS environment. The mechanical requirement for a seamless user experience allowed data across the infrastructure to be saved and retrieved by WoodRock employees, which maintained the cleanliness and security of the network environment. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *File Explorer* in *Windows Taskbar* window | *This PC* | *Local Disk (C:)* | *DFS*
2. Right-clicked *RoamingProfiles*
3. Clicked *Properties* | *Sharing* | **Advanced Sharing** in *Sharing* window | *Share this folder*
4. Typed `RoamingProfiles$` in *Share Name* field
5. Clicked **Permissions** in *Advanced Sharing* | *Everyone*
6. Selected *Allow Full Control*
7. Clicked **OK** in *Share Permissions* window | **OK** in *Advanced Sharing* window | **Close** in *RoamingProfiles Properties* window
8. Right-clicked *RedirectedFolders*
9. Clicked *Properties* | *Sharing* | **Advanced Sharing** in *Sharing* window | *Share this folder*
10. Typed `RedirectedFolders$` in *Share Name* field
11. Clicked **Permissions** in *Advanced Sharing* window | *Everyone*
12. Selected *Allow Full Control*
13. Clicked **OK** in *Share Permissions* window | **OK** in *Advanced Sharing* window | **Close** in *RoamingProfiles Properties* window

Folder Targets Addition on G03SRV01VM

The section detailed how the logical names were mapped to the physical network paths, which was the last part of the DFS configuration. For WoodRock, the steps tied the user-friendly namespace paths to the invisible administrative shares created earlier. The folder targets addition completed the business's abstraction of WoodRock's storage architecture, which allowed users and system services to simply interact with `\\group03.c24200.cit.lc\DFS\RedirectedFolders` without the knowledge of which server actually hosted the data at any given time. The setup was a mechanical necessity for high availability, which allowed administrators to switch backend

storage hardware in the future without the need of the file paths (used by employees or Group Policy settings) being changed. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* | *Tools* | *DFS Management* | *\\group03.c24200.cit.lcl\DFS* | *New Folder*
2. Typed `RedirectedFolders` in *Name* field
3. Clicked **Add** in *New Folder* field
4. Typed `\\G03SRV01VM\RedirectedFolders$` in *Path to folder target* field
5. Clicked **OK** in *Add Folder Target* window | **OK** in *New Folder* window | *New Folder* in *Actions* window
6. Inputted `RoamingProfiles` in *Name* field
7. Clicked **Add** in *New Folder* field
8. Typed `\\G03SRV01VM\RoamingProfiles$` in *Path to folder target* field
9. Clicked **OK** in *Add Folder Target* window | **OK** in *New Folder* window

New Replication Group Creation on G03SRV01VM

The section detailed how a Multipurpose Replication Group was configured for data synchronization between G03SRV01VM and G03SRV04. For WoodRock, the configuration was the foundation of the company's disaster recovery plan, which ensured a live, identical copy of all user documents and profiles was maintained on a secondary server. As G03SRV01VM was designated as the primary DC where the `RedirectedFolders` and `RoamingProfiles` directories were stored and copied to G03SRV04, the company eliminated the risk of a "single point of failure." The redundant architecture was essential for high availability. In the event of the primary DC failure, WoodRock employees maintained access to the files from the secondary

host without any data loss or work interruption. **Refer to Appendix B: System and Network Configuration.**

1. Selected *Server Manager* | *Tools* | *DFS Management*
2. Clicked *New Replication Group* | **Next** in *Replication Group Type* window
3. Typed *Multipurpose Replication Group* in *Name of replication group* field
4. Clicked **Next** in *Name and Domain* window | **Add** in *Replication Group Members* window
5. Typed *G03SRV04* in *enter the object names to select* field
6. Clicked **Check Names** | **OK** in *Selection Computers* window | **Next** in *Replication Group Members* window | **Next** in *Topology Selection* window | **Next** in *Replication Group Schedule* window
7. Selected *G03SRV01VM* as *Primary Member*
8. Clicked **Next** in *Primary Member* window | **Add** in *Folders to Replication* window | **Browse** | *DFS* | *RedirectedFolders* | **OK** | **OK** in *Add Folder to Replicate* window | **Add** | **Browse** | *DFS* | *RoamingProfiles* | **OK** | **Next** in *Folders to Replicate*
9. Selected **Next** in *Local Path* window | **Next** in *Local Path* window | **Create** in *Review Settings* window | **Close** in *Confirmation*

GPOs Configuration for DFS on G03SRV01VM

The section detailed how certain GPOs were set up which automatically sent user data to the DFS namespace. For WoodRock, the process of having moved user environments from local hardware to the secure replicated storage cloud was critical. As the *RedirectedFolders* and *RoamingProfiles* paths were set up through GPO, WoodRock made sure every employee, no matter which workstation was used, was automatically connected to the personal files and

desktop settings. The centralized management was necessary for WoodRock's business operation efficiency where the need for manual data migrations was eliminated and the WoodRock security and backup policies were applied consistently to all user-generated content across the domain.

Refer to Appendix B: System and Network Configuration.

1. Clicked *Server Manager* | *Tools* | *Group Policy Management* | *Domains* |
group03.c24200.cit.lcl
2. Right-clicked *Folder Redirection*
3. Clicked *Edit* | *User Configuration* | *Policies* | *Windows Settings* | *Folder Redirection*
4. Right-clicked *Desktop*
5. Selected *Properties* | *Redirect to the following location* dropdown
6. Typed `\\group03.c24200.cit.lcl\DFS\RedirectedFolders$` in *Root*
field
7. Clicked **OK** in *Desktop Properties* | **Yes** in *Warning*
8. Right-clicked *Roaming Profiles*
9. Clicked *Edit* | *Computer Configuration* | *Policies* | *Administrative Templates* | *System* |
User Profiles
10. Right-clicked *Set roaming profile path for all users logging onto this computer*
11. Clicked *Edit*
12. Typed
`\\group03.c24200.cit.lcl\DFS\RoamingProfiles$\%USERNAME%` in
Users logging onto this computer should use this roaming profile path
13. Clicked **OK** in *Set Roaming profile path for all users logging onto this computer* window

Print Operator/Administrator User Creation on G03SRV01VM

The section detailed how a special administrative user account was created and added to the Print Operators security group. For WoodRock, the process followed the principle of least privilege where specific environmental management tasks were delegated to someone else without the need of full Domain Administrator rights. A dedicated operator ensured print services and hardware queues were well managed to support day-to-day office work while a high level of security was maintained. The administrative setup was important for smooth operations where printing and resource management problems were allowed to be resolved quickly across the domain while the WoodRock directory services were kept secure. **Refer to Appendix B: System and Network Configuration.**

1. Selected *Server Manager* in *Windows Taskbar* | *Tools* | *AD Users and Computers*
2. Right-clicked *group03.c24200.cit.lcl*
3. Selected *New* | *User*
4. Typed *Hannah* in *First name* field
5. Inputted *Doll* in *Last Name* field
6. Typed *hdoll* in *User logon name* field
7. Clicked **Next** in *New Object - User* window
8. Typed *password* in *Password* field
9. Inputted *password* in *Confirm Password* field
10. Selected **Next** in *New Object - User* window | **Finish** in *New Object - User* window |
Builtin folder in *Active Directory Users and Computers* window
11. Right-clicked *Print Operators*
12. Selected *Properties* | *Members* tab | **Add...** in *Print Operators Properties* window

13. Typed `hdoll` in *Enter the object names to select* field
14. Clicked **Check Names** in *Select Users, Contacts, Computers, Service Accounts, or Groups* window | **OK** in *Select Users, Contacts, Computers, Service Accounts, or Groups* window | **OK** in *Print Operators Properties* window
15. Selected **Close** in *Active Directory Users and Computers* window

Printer Installation on G03SRV01VM

The section detailed how the installation of the Print and Document Services role and the configuration of the network printing hardware were incorporated into the domain environment. For WoodRock, the centralization of the print management on the DC was a mechanical requirement where employees discovered and utilized office hardware without manual configuration on each workstation. As the Internet Printing Client was enabled and added to the AD directory, WoodRock made things easy for all employees where the Hewlett-Packard (HP) LaserJet E40040 could be found and used. The setup was necessary for smooth operations where administrators managed print queues, updated drivers, and enforced usage policies all from one place, which cut down on the technical overhead as office peripherals spread out. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Windows Taskbar* | *Local Server* | *Manage* | *Add Roles and Features* | **Next** in *Before You Begin* window | **Next** in *Installation Type* window | **Next** in *Server Selection* window | **Next** in *Server Roles* window
2. Selected *Internet Printing Client* | **Next** in *Features* window | **Install** in *Confirmation* window | **Close** in *Results* window
3. Clicked *Manage* in *Server Manager* window | *Add Roles and Features* | **Next** in *Before You Begin* window | **Next** in *Installation Type* window | **Next** in *Server Selection* window

4. Selected *Print and Document Services* | **Add Features** | **Next** in *Server Roles* window | **Next** in *Features* window | **Next** in *Print and Document Service* window | **Next** in *Role Services* window | **Install** in *Confirmation* window | **Close** in *Results* window
5. Clicked *Tools* in *Server Manager* window | *Print Management* | *Printer Servers*
6. Right-clicked *G03SRV01VM (local)*
7. Selected *Add Printer...* | **Next** in *Printer Installation* | *Type of Device* dropdown | *Autodetect*
8. Typed 44.3.1.238 in *Host name or IP address* field
9. Selected **Next** in *Printer Address* window | **Next** in *Printer Name and Sharing Settings* window | **Next** in *Printer Found* window | **Finish** in *Completing the Network Printer Installation Wizard* window | *G03SRV01VM* in *Print Management* window | *Printers*
10. Double-clicked *k238 (HP LaserJet E40040)*
11. Selected *Sharing* tab | *List in the directory* | **Apply** | **OK** in *k238 (HP LaserJet E40040) Properties* window | **Close** in *Print Management* window

Printer GPOs Setup in G03SRV01VM

The section detailed how a GPO was implemented for printer deployment across the AD domain. Printers assignment on every single workstation for WoodRock was neither efficient nor practical to. However, the use of Group Policy Preferences enabled the HP LaserJet E40040 to be pushed to every user profile after the same user had logged on. The GPO was a mechanical requirement for the business continuity where employees had quick access to critical hardware resources, regardless of which Windows Person Computers (PCs) the employees used. “gpupdate /force” command was run so policy was forcefully updated where changes were made

immediately and employees maintained high productivity and a seamless experience throughout the entire office infrastructure. **Refer to Appendix B: System and Network Configuration.**

1. Selected *Server Manager* in *Windows Taskbar* | *Tools* | *Group Policy Management*
2. Right-clicked *Default Domain Policy*
3. Selected *Edit...* | *Computer Configuration* | *Policies* | *Administrative Template* | *Printers*
4. Double-clicked *Allow printers to be published*
5. Selected *Enabled* | **Apply** | **OK** in *Allow printer to be published* window | *User Configuration* | *Preferences* | *Control Panel Settings*
6. Right-clicked *Printers*
7. Selected *New* | *Shared Printer* | ... in *Share path* field
8. Double-clicked *G03SRV01VM-k238 (HP LaserJet E40040)*
9. Selected **Apply** | **OK** in *New Shared Printer Properties* window | **Close** in *Group Policy Management Editor* window | **Close** in *Group Policy Management* window | *Powershell* in *Windows Taskbar*
10. Typed `gpupdate /force` in *C:\WINDOWS\system32* field
11. Pressed **Enter**
12. Typed `Y` in *OK to log off? (Y/N)* field
13. Pressed **Enter**

Second Duplex Queue Addition for k238 on G03SRV01VM

The section detailed the configuration of a secondary, specialized print queue which was designed specifically for double-sided printing. For WoodRock, the addition of a dedicated k238-duplex queue was a strategic move where resource conservation was encouraged and operating costs associated with paper consumption was reduced. As a separate logical printer

object was established with the Duplex Unit pre-installed and activated, the company improved the user experience where employees selected a particular “Duplex” option directly from the print menu rather than digged through driver settings. The configuration was a mechanical requirement for sustainable office practices where the infrastructure was flexible enough for various document formatting needs to be met. **Refer to Appendix B: System and Network Configuration.**

1. Selected *Server Manager* in *Windows Taskbar* | *Tools* | *Print Management*
2. Right-clicked *Printers*
3. Selected *Add Printer* | **Next** in *Printer Installation* window | *TCP/IP Device* in *Type of Device* dropdown
4. Typed 44.38.1.238 in *Host name or IP address* field
5. Clicked **Next** in *Printer Address* window | **Next** in *Additional port information required* window | *Use an existing print driver on the computer* | **Next** in *Print Driver* window
6. Typed k238-duplex in *Printer Name* field
7. Selected **Next** in *Printer Name and Sharing Settings* window | **Next** in *Printer Found* window | **Finish** in *Network Printer Installation Wizard* window
8. Right-clicked *k238-duplex*
9. Clicked *Properties* | *Device Settings* | *Duplex Unit* dropdown
10. Selected *Installed*
11. Clicked **OK** in *k238-duplex Properties* window

Printing Pause Configuration for Print Queues on G03SRV01VM

The section detailed how the print queues were suspended with administrative privilege for both the standard and duplex versions of the k238 printer. For WoodRock, the suspension of

printing was a mechanical requirement for routine hardware maintenance, paper jams cleanup, or large print jobs management without compromise of incoming print jobs. As the queues were suspended at the server level on G03SRV01VM, WoodRock ensured user submissions were safely held in the printing spooler, rather than failed or disappeared during a service window. The control was vital to WoodRock's operational reliability where the administrators stabilized the printing environment prior to resumption of full production, thus wasted resources were avoided and document integrity was ensured. **Refer to Appendix B: System and Network**

Configuration.

1. Selected *Server Manager* in *Windows Taskbar | Tools | Print Management*
2. Right-clicked *k238 (HP LaserJet E40040)*
3. Selected *Open Printer Queue | Printer | Pause Printing | Close* in *k238 (HP LaserJet E40040)* window
4. Right-clicked *k238-duplex*
5. Selected *Open Printer Queue | Printer | Pause Printing*

Printer Permissions Configuration for Print Queues on G03SRV01VM

The section detailed how the application of granular access controls was applied to enterprise print queues. For WoodRock, the configuration created a tiered usage policy which followed sustainable printing practices where access was limited to hardware selection capabilities. As the “Everyone” group was assigned to the k238-duplex queue and general permissions were removed on the standard k238 queue, the company effectively forced all users into double-sided print mode by default. The printer permissions configuration measure was critical for operational expenses management and security enhancement where large-scale print jobs utilized optimal settings and comprehensive administrative privileges were reserved for

approved individuals like Print Operators. The configuration safeguarded hardware assets and promoted prudent use where unauthorized alterations were prevented. **Refer to Appendix B:**

System and Network Configuration.

1. Clicked *Server Manager* in *Windows Taskbar | Tools | Print Management*
2. Right-clicked *k238-duplex*
3. Clicked *Properties | Security* tab | *Everyone*
4. Selected *Print*
5. Clicked **OK** in *k238-duplex Properties* window
6. Right-clicked *k238 (HP LaserJet E40040)*
7. Clicked *Properties | Security* tab
8. Selected *SYSTEM*
9. Clicked **Remove** in *Group or user names* window
10. Repeated steps 8-9 for all users and groups other than Administrator, Administrators, Print Operators, and CREATOR OWNER
11. Clicked **OK** in *k238 (HP LaserJet E40040) Properties* window

Printer Priority Configuration for Print Operator and Admins on G03SRV01VM

The section detailed how job priority could be set for the main print queue. For WoodRock, printer priority was a strategic necessity where time-sensitive administrative documents were processed before normal user jobs when network traffic was high. As the priority value of the k238 logical printer object (non-duplex) was increased to 99, the company ensured print jobs were initiated by authorized personnel such as the Print Operator or Domain Administrators which then transferred to the hardware spooler. The optimization was important for WoodRock's efficiency, where critical business reports or security documentation were not

delayed by high-volume, non-essential print jobs, which kept the organization's leadership responsive and productive. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Server Manager* in *Windows Taskbar* | *Tools* | *Print Management*
2. Right-clicked *k238 (HP LaserJet E40040)*
3. Clicked *Properties* | *Advanced* tab
4. Typed 99 in *Priority* field
5. Clicked **OK** in *k238 (HP LaserJet E40040) Properties* window

PowerShell Remoting Configuration on all VMs

The section detailed how PowerShell Remoting was configured and tested within the enterprise infrastructure. For WoodRock, the ability of the commands execution and configuration management on multiple VMs from a single terminal was a mechanical necessity for scalable administration and rapid incident response. As the WinRM was configured with the use of the `Enable-PSRemoting` command, WoodRock moved away from time-consuming, per-machine manual logins to a centralized, scriptable management model. The Powershell Remoting setup was a key enabler of WoodRock's operation agility where administrators were allowed to perform critical maintenance tasks such as auditing, stopping and starting services (like the Print Spooler) on remote hosts such as G03SRV03VM, from the primary workstation. The setup ensured consistent uptime and security across the entire WoodRock server cluster.

Refer to Appendix B: System and Network Configuration.

1. Right-clicked *PowerShell* in *Windows Taskbar*
2. Clicked *Run as administrator*
3. Typed `Enable-PSRemoting -Force` in *C:\Users\System32* field
4. Pressed **Enter**
5. Typed `Enter-PSSession -ComputerName G03SRV03VM -Credential Group03\Administrator` in *C:\Users\System32* field
6. Pressed **Enter**
7. Typed password in *Password* field
8. Clicked **OK** in *Windows PowerShell credential request* window
9. Typed `Get-Service; Stop-Service -Name Spooler` in *C:\Users\System32* field

10. Pressed **Enter**

11. Typed `Get-Service -Name Spooler;Start-Service -Name Spooler` in `C:\Users\System32` field

12. Pressed **Enter**

13. Inputted `Exit-PSSession` in `C:\Users\System32` field

14. Pressed **Enter**

15. Repeated steps 1-14 for all VMs in Prism Central

WAC Installation on G03SRV06

The section detailed how WAC, a modern, browser-based management platform which brought together all of the tools needed to manage servers into one interface, was set up. For WoodRock, the use of WAC was a strategic move which replaced old consoles with a single dashboard. The WAC integration was made possible by the WinRM exceptions and PowerShell Remoting enabled in the previous phase, which served as the underlying communication channel where all servers and Windows PCs across the domain were connected and managed by WAC. As a self-signed certificate was automatically generated and pre-configured exceptions were used, the entire fleet could be monitored and configured by WoodRock's administrators from any workstation. The installation was a mechanical necessity for modernized operations which streamlined complex tasks (such as certificate management and hardware monitoring) into a single, efficient workflow (which enhanced the overall security and agility of the WoodRock Information Technology infrastructure). **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Microsoft Edge* in *Windows Taskbar*

2. Typed `https://aka.ms/WACDownload` in *Search bar* field

3. Pressed **Enter**
4. Hit **Ctrl + J**
5. Clicked *Open file* in *WindowsAdminCenter2511.exe* field | **Next** in *Windows Admin Center (v2) Installer* window | *I accept these terms and understand the privacy statement* | **Next** in *License Terms and Privacy Statement* window
6. Selected **Next** in *Select installation mode* window | *Generate a self-signed certificate* | **Next** in *Select TLS certificate* window | **Next** in *Automatic updates* window | **Next** in *Send diagnostic data to Microsoft* window | **Install** in *Ready to Install* window
7. Clicked **Finish** in *Completing the Windows Admin Center (v2) Setup Wizard* window | *Windows Admin Center (v2)* in *Windows Taskbar* | **Advanced** in *Microsoft Edge* browser | *Continue to g03srv06.group03.c24200.cit.lcl (unsafe)*
8. Typed `GROUP03\Administrator` in *Username* field
9. Inputted password in *Password* field
10. Pressed **Enter**

Windows Server 2025 and Windows 11 Addition on WAC

The section detailed how WAC inventory was populated with the organization's Windows Server 2025 and Windows 11 VMs. For WoodRock, the process transformed a stand-alone management tool into a comprehensive command center for the entire domain. Using the AD search capability and the WinRM connectivity established earlier, administrators brought both server-side infrastructure and client-side workstations under a single window panel. The centralized visibility was a mechanical necessity for proactive maintenance which enabled system health simultaneous monitor, resource utilization, and status across patches distribution in both server and desktop environments. The integration ensured the WoodRock Information

Technology team managed the wide range of Nutanix-hosted fleets with the least amount of work and the most efficiency. **Refer to Appendix B: System and Network Configuration.**

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Typed `https://G03SRV06` in *Search bar* field
3. Pressed **Enter**
4. Typed `GROUP03\Administrator` in *Username* field
5. Inputted password in *Password* field
6. Pressed **Enter**
7. Clicked + **Add** in *Windows Admin Center* | + **Add manually** | **Add** in *Servers* window |
Search Active Directory
8. Typed `G03SRV01VM.group03.c24200.cit.lcl` in *Search* field
9. Clicked **Add** in *Server* window
10. Repeated steps 7-9 for all Windows Server 2025 VMs in Nutanix Prism Central
11. Clicked + **Add** in *Windows Admin Center* | + **Add manually** | **Add** in *Windows PCs*
window | *Search Active Directory*
12. Typed `G03S1WIN11VM.group03.c24200.cit.lcl` in *Search* field
13. Clicked **Add** in *Windows PC* window
14. Repeated steps 11-13 for all Windows 11 VMs in Nutanix Prism Central

RESULTS

The WoodRock enterprise environment was further expanded through the addition of Windows Server Update Services, distributed file system under a namespace, printing, remote administration, and a centralized windows admin center. The newly implemented services expanded the previous multi-site infrastructure by improving operational efficiency and security.

The addition of WSUS on G03SRV07 allowed for a centralized system to approve and apply updates for all systems in the WoodRock domain. Through group policy, client machines and servers were configured for updates to be received from the WSUS server rather than the default Microsoft servers. G03SRV07 received a 500GB storage device for updating content of Windows Server 2025 and Windows 11 devices. Synchronization with Microsoft Update was successful, and testing confirmed the domain joined machines were registered with WSUS and received approved updates. The implementation of WSUS allowed for consistent deployment of updates, which reduced the need for administrators to manually update machines and allowed for a more secure, automated environment.

The Distributed File System was successfully integrated into the WoodRock environment to create a unified namespace for client file access. Shared folders for roaming profiles and redirected folders were created and mapped through the DFS namespace, which allowed users to access data through a single logical path regardless of site or server being used. Replication between G03SRV01VM and G03SRV04 was configured and verified using DFS Management and targeted folders, which ensured redundancy across sites. Logon through client machines and different users confirmed the shared folders functioned correctly and maintained a consistent environment across the domain.

ENTERPRISE WINDOWS SERVER ADMINISTRATION

Enterprise printing services were implemented on G03SRV01VM, which was configured by deploying the HP LaserJet E40040 printer to be accessed throughout the domain. Along with the LaserJet, a duplex print queue was configured to enable printing for general users, with the addition of allowing administrator users to access a higher priority print queue. Using Active Directory groups, permissions were configured to allow print operators and administrators to have certain control over print jobs and queues. Testing through normal and administrative users showed print priority was enforced correctly, and users could access printers through group policy.

PowerShell remoting was enabled across all servers to allow administrators to remotely connect and manage services running on other servers, which could be executed through the use of commands in the terminal. Testing remote session connections showed successful connectivity between servers and execution of commands to start and stop services functioned correctly. By implementing remote connection, administrators no longer needed direct access to a server.

Finally, the deployment of Windows Admin Center on G03SRV06 provided a centralized interface for monitoring and accessing machines in the domain. Windows Server 2025 and Windows 11 machines were added to the WAC dashboard using WinRM configurations. Through the addition of the machines to the WAC, administrators could remotely manage domain machine services, review event logs, and monitor system health. Testing showed the ability to connect to systems through the WAC and consistent access to all domain machines. By implementing the WAC, all systems could be accessed through a single interface to increase administrative efficiency and control within the WoodRock enterprise environment.

The newly improved enterprise environment was able to operate without errors following proper configuration. DFS replication was consistent across both sites, WSUS was able to

provide proper updates to all machines, and printer services worked accordingly, along with the set permissions. The addition of the new services improved WoodRock's internal environment by allowing for fewer manual administrative services and providing a more functional system, which is aligned accordingly with WoodRock's business scenario.

Figure 3: Logical Model

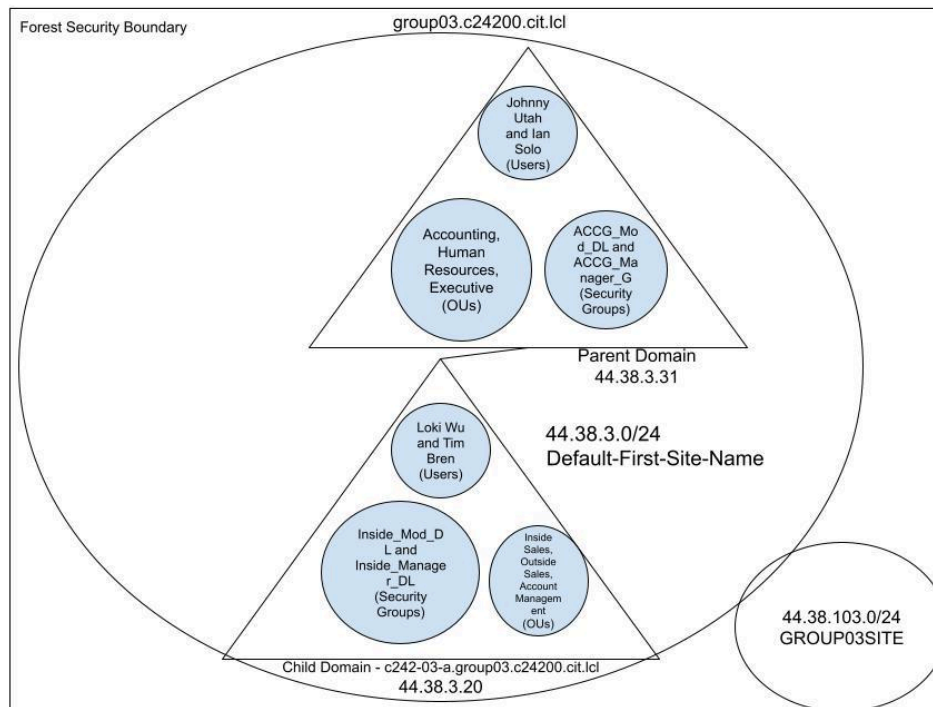


Figure 3.) Logical model of WoodRock's forest

WoodRock decided to maintain the overarching forest topology with little changes having been undergone. The two domains, child and parent, held personal department OUs, which were Accounting, Human Resources, and Executive for the parent. In the child domain, Inside Sales, Outside Sales, and Account Management were the OUs hosted. Users Johnny Utah and Ian Solo were once again atop the competition within the Accounting department, and were admitted to the ACCG_Mod_DL security group. The child domain brought users Loki Wu and Tim Bren success once more, which granted the right to the Inside_Manager_DL security group.

ENTERPRISE WINDOWS SERVER ADMINISTRATION

Replication occurred between the two sites in the parent domain and GROUP03SITE held a separate subnet range in 44.38.103.0/24 with no resided domain.

Figure 4: Physical Model

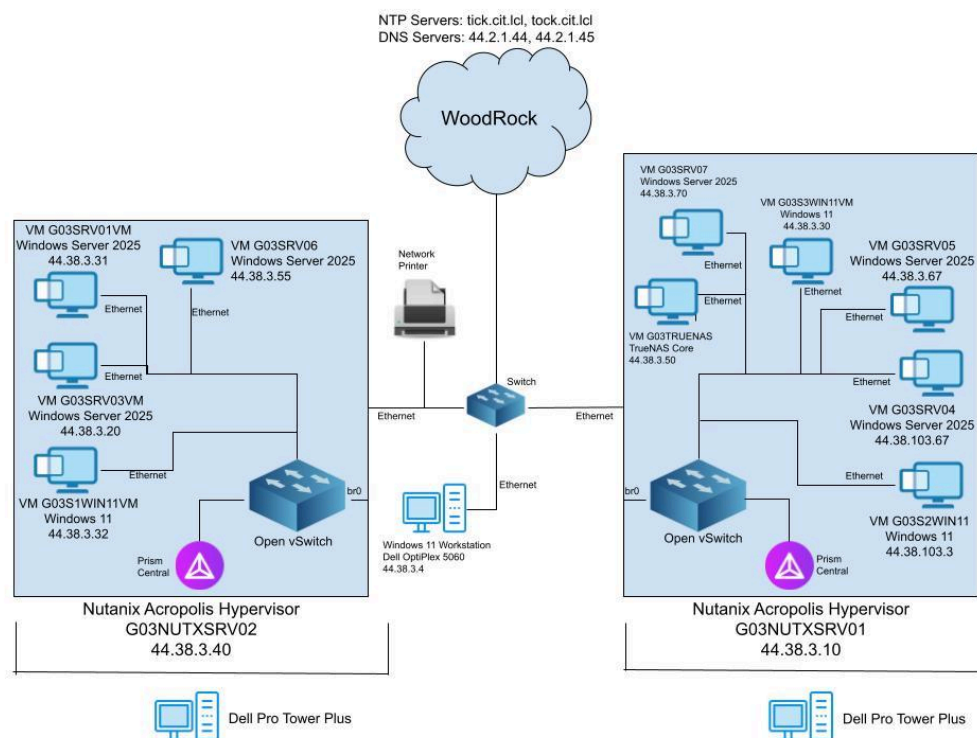


Figure 4.) Physical model of WoodRock topology

WoodRock's topology was developed further with the addition of another VM called G03SRV07, which booted a Windows Server 2025 image at 44.38.3.70. The additional VM was hosted on the G03NUTXSRV01 cluster, and was connected to the Open vSwitch via ethernet in the same manner as the other VMs, which pointed to Prism Central. The two clusters were held on two AHVs hosted by separate Dell Pro Tower Plus machines. The hypervisors were connected to the Network Printer, Dell OptiPlex 5060, and the WoodRock CIT network through ethernet. WoodRock's company network printer had an additional print queue added to manage priority and traffic on the printer.

CONCLUSIONS AND RECOMMENDATIONS

The completion of the expansion of WoodRock's environment converted WoodRock's infrastructure from a single-site domain to a multi-site environment with centralized management, access control, redundancy, and scalability. Using centralized update management, distributed file systems, enterprise printing, and remote administration, the organization's security, efficiency, and reliability have greatly improved.

Every newly implemented service improves the infrastructure of WoodRock. Active Directory allowed for a functional domain with proper configurations. WSUS ensured consistent patch deployments, which reduced the need for administrator intervention. DFS namespaces and replication provided users with consistent and easy to understand data across the sites. Printing services improved resource usage and management, while PowerShell remoting increased administrative capabilities by improving the ability of response time through remote management.

Combined, the newly implemented services provided an environment which improved upon issues addressed in the business scenario. WoodRock's improved infrastructure demonstrates proper configurations, including replication across sites, proper access control and authentication, consistent patches, and administrative tools. The original objectives stated in the business scenario were accomplished, resulting in an enterprise environment capable of meeting user needs.

RECOMMENDATIONS

The following recommendations detailed the lessons learned reflection and advice during the configuration of the multi-site enterprise environment for WoodRock. Recommendations were intended for future administrators who could have built an environment similar to WoodRock where certain actions or steps were shown for the efficiency improvement of configurations and the mitigation of problems troubleshooting. Challenges faced during the configuration of the WoodRock environment were mentioned and explained in assistance of future administrators with implementation.

Recommendation 1: Separate Group Policy Objects by function rather than grouping the policies within the default domain policy. GPOs were used to configure WSUS and printer services, however, when troubleshooting the GPOs given to the services, the problems were difficult to identify and sort through if all the policies were grouped within the default domain policy. Instead of relying on the default domain policy, dedicated GPOs should be created and enforced in the domain and linked to specific organizational units to allow for a more fine grained environment. With a more fine grained GPO environment, visibility for troubleshooting had significantly improved, and changes to one GPO was less likely to inadvertently affect other services within the domain.

Recommendation 2: WinRM and firewall configurations should have been verified on all VMs before WAC was implemented. Default WinRM firewall rules were too restrictive for the machines, which operated from the second site, to be implemented. As the proper rules were manually configured and Enable-PSRemoting was run on all machines before attempting to connect to WAC, issues could be avoided. Later troubleshooting could be

avoided during the final configurations of WAC with the validation of the firewall rule which allowed remote access early on.

Recommendation 3: Before an image of a virtual machine was created as a template, template systems should have been configured with sysprep for Security Identifiers duplication to be avoided. If VMs shared system identifiers, conflicts arose with services such as WSUS, which prevented proper policy application and update management. Images used as templates for future virtual machines required correct configuration so the VMs properly functioned in the future. Before an image was created, the system should be generalized; if the previously mentioned process could be ensured early on in the configuration of an enterprise environment such as WoodRock, troubleshooting would be avoided, and identity dependent services would have functioned as intended.

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APPENDIX A: PROBLEM SOLVING

The problem solving section provided a comprehensive overview of problems faced during the configuration of all systems and virtual networks in the Server Administration of the systems. The first problem faced involved significant WinRM connectivity barriers which prevented remote hosts from being managed by WAC, particularly the G03SRV04 server located at secondary sites and the Windows 11 PCs. The second problem faced in the Enterprise Microsoft Windows Server Administration process happened when Windows 11 client workstations were unable to be connected to shared printers due to restrictive default security descriptors on the print server. The third problem faced was the accidental closure of the WSUS Post-Installation Configuration Wizard, which required a manual restoration of the setup interface through the management console options. As the challenges and the respective solutions were documented, the problem solving section served as a technical knowledge base for the integrity and connectivity of the WoodRock infrastructure to be maintained.

Problem 1: WinRM Connectivity for Remote Management in WAC

Problem Description: After the VMs were populated in WAC, a problem happened with remote connections, especially with the Windows 11 VMs and the G03SRV04 server. Specifically, WAC would show "Connection Refused" or "WinRM cannot process the request" errors every time remote connection establishment between WAC and VMs from Nutanix Prism Central was attempted, even after the WAC portal was refreshed several times. The problem was not anticipated as PowerShell Remoting was expected to be enabled on all machines prior to the connection process. Some servers eventually connected after a few manual retries, but the Windows 11 VMs and the secondary-site server, G03SRV04, were still completely unreachable for remote management.

Problem Solutions: In order for the communication barriers to be overcome, the WinRM

listener needs to be explicitly enabled and the corresponding firewall exceptions need to be active. As the “Enable-PSRemoting” command was executed on every Windows 11 VMs, the system, expectedly, automatically configured the WinRM service where the necessary firewall exception rules for WinRM incoming traffic was created and started. Furthermore, G03SRV04 server, which resided on different subnets or secondary sites, was hypothesized the firewall scope might be too restrictive; hence, G03SRV04 server required a manual adjustment of the “WINRM-HTTP-In-TCP-PUBLIC” rule, which enabled traffic from the WinRM remote address to be eventually accepted on G03SRV04.

Solutions Attempted: The first approach was the execution of “Enable-PSRemoting” command in the PowerShell on all the VMs with the expectation where the standard configuration would be enforced. The first approach successfully brought the Windows 11 VMs online in WAC but failed for the Windows Servers (G03SRV04 in particular), which showed no logs and remained disconnected. Initially, a network routing problem was suspected since G03SRV04 was at the second site. However, basic connectivity was confirmed by a successful ping request, which confirmed the host was reachable. After the documentation provided by the Microsoft “about_Remote_Troubleshooting” guide was reviewed, the problem was found where the traffic from GROUP03SITE subnet was being specifically blocked by the default WinRM firewall rules on G03SRV04 server. As the final solution was tested, the administrators applied the firewall rule adjustment on G03SRV04 and immediately refreshed the connection status in WAC; as a result, the WAC connection status shifted from "Connecting..." to "Online" instantly, and the

administrators, who logged into G03SRV04 as a standard user, successfully confirmed the link was stable.

Final Solution: Specific PowerShell commands were used based on the operating system and site location. For the Windows 11 workstations, “Enable-PSRemoting” was executed to make the necessary firewall exceptions for WAC integration. For the G03SRV04 server, the firewall scope had to be manually expanded with the use of “Set-NetFirewallRule -Name "WINRM-HTTP-In-TCP-PUBLIC" -RemoteAddress Any” command in an elevated PowerShell session which required administrator’s privilege. The solution provided by Microsoft guide enabled management packets from the primary site to be accepted by the G03SRV04 server which resided on the GROUPO3SITE site. The other Windows Servers didn't need any extra configuration because the correct settings were inherited from the AD domain's default site.

Problem 2: "Access Denied" Errors with Windows 11 Printer Connection

Problem Description: Windows 11 workstations were unable to be connected to the shared network printer, and the standard user, “ckent”, received the error message, “Windows Cannot Connect to the Printer. Access Is Denied.” The printer, which was selectable by the user “ckent”, was listed in the user directory. However, the connection handshake failed immediately after the “Connect” button was clicked from the user’s point of view. The problem appeared to be a Windows 11 specific problem as the connection worked on Windows Server units with the use of accounts where Print Operator rights were granted. The listed facts indicated a change in user rights.

Problem Solutions A solution was proposed where the printer's security descriptors were aligned with what standard AD users expected. As a solution, the necessary “Print”

permissions were granted to standard users who normally hadn't gotten. Hence, "Everyone" group was added to the printer's security list with the use of the Print Management tool in Server Manager on the G03SRV01VM server. The solution expectantly solved the problem as "Access Is Denied" error was a classic indicator of a permission mismatch between the client's credentials and the print server's Access Control List (ACL). The ACL ensured the server authenticated the user's request rather than rejected the printer connection request at the gateway.

Solutions Attempted: At first, Group Policy and local rights were looked for the driver installation, instead of the printer object. First, the existing "Printer" GPO from a previous administration was applied across the AD domain with the use of the command "gpupdate /force", which made sure the GPO was active. However, the printer connection still couldn't be established from one of the Windows 11 VMs. The second attempt was also involved with the use of GPO where the "Everyone" group was added to the "Load and unload device drivers" policy and the "Disallow installation of printers using kernel-mode drivers" setting was set to "Disable". As the administrators thought of a driver-level block, the second attempt failed because the problem wasn't the client's ability to install a driver, but the print server explicitly denied the user's request for access to the print queue. After the final solution was tested, security changes were applied on the server, and the G03S1WIN11VM was logged into with a standard user "ckent"; the administrators successfully connected to the printer and verified the fix with a successful test page print request sent to the queue.

Final Solution: The final solution was implemented on G03SRV01VM where the server-side permissions were ensured to be correctly broadcast. First, the administrators navigated to

Server Manager, clicked on Print Management from the Tools panel, and found the k238 printer under the Printers node. Then, the administrators right-clicked on the printer, navigated to the Security tab, added the “Everyone” group, and selected the Print permission which allowed standard domain users printing permission. After the change was made, the Windows 11 VMs successfully connected to the printer and used the printers without any more help from the administrator.

Problem 3: Disappearance of the WSUS Post-Installation Configuration Wizard

Problem Description: During the initial setup of WSUS on G03SRV07, a serious procedural mistake was made when the administrators moved from installation to configuration. The role of WSUS was successfully added using the “Add Roles and Features” wizard in Server Manager, and the post-installation configuration wizard was launched automatically as expected. Once launched, the administrative team, who took a break and recorded the steps of the setup process, inadvertently closed the wizard which resulted in the disappearance of the configuration interface.

Problem Solutions: Administrators sought for a method where the configuration process could be manually re-triggered without reinstallation of the entire WSUS role, which could be time-consuming and inefficient. The administrative team identified an obvious fact: while the wizard was a “post-installation” task, the wizard was essentially a component of the WSUS Management Console rather than a one-time system event. Hence, navigation through the WSUS console options, expectantly, worked because the management architecture was designed where the initial setup parameters could be re-run at any time by the administrator who then modified upstream server settings or synchronization schedules.

Solutions Attempted: The administrators first looked for a “Resume” or “Flag” notification in the Server Manager notification flag (the “Tasks” menu), but even though notification for “Post-deployment configuration” appeared, the full interactive wizard couldn’t be brought back with a simple click. Multiple tries were attempted, but a standalone executable for the wizard in the system folders couldn’t be found because the wizard was part of the Microsoft Management Console (MMC) snap-in. As the final solution was tested, the administrative team went through the WSUS management tree to the specific “Options” node and selected the configuration wizard. The administrators knew the final solution had worked when the original setup window came back where the administrative team continued the synchronization setting configuration.

Final Solution: The final solution involved a targeted navigation path through the specialized management tools installed with the role. The administrators opened WSUS from the Tools menu in Server Manager and expanded the console tree: Update Services | G03SRV07 | Options. From the list of available settings, the administrative team selected WSUS Server Configuration Wizard, which immediately re-launched the interface. From here, the team continued with the upstream update source selection, proxy setting configuration, and the first sync initiation with Microsoft Update.

APPENDIX B: SYSTEM CONFIGURATION

The section provided an overview of the configurations of all systems deployed and altered during the configuration of the enterprise Windows Server administration environment. The following tables outlined the software specification of each system and recorded how each system was organized for management. All newly created VMs, WSUS configurations, DFS namespaces, and administration tools were outlined below.

Table 1: VM Configurations

VM configurations and respective roles

VM Name	IP Address	Role
G03SRV01VM	44.38.3.31	Primary domain controller, primary DFS namespace server,
G03SRV04	44.38.103.2	DFS namespace replication server
G03SRV06	44.38.3.55	Windows Admin Center host
G03SRV07	44.38.3.70	WSUS server
G03S1WIN11VM	44.38.3.32	Client machine

Table 2: Service Placements

Service placement configurations defining each server-service placement

Service	VM Name	Role
Windows Server Update Services	G03SRV07	Centralized distribution of OS updates
Distributed File System Namespace and Replication	G03SRV01VM/G03SRV04	Unified folder paths and replication for redirected folders and roaming profiles
Printing	G03SRV01VM	Print queue control with priority control

Windows Admin Center	G03SRV06	Single interface for managing all windows servers and clients
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Table 3: Shared Folder Configurations

All file paths used for DFS namespace and replication

Server	Folder Path	Share Name	Role
G03SRV01VM	C:\DFS\RedirectedFolders	RedirectedFolders\$	Storage for redirected folders
G03SRV01VM	C:\DFS\RoamingProfiles	RoamingProfiles\$	Storage for roaming profiles
G03SRV04	C:\DFS\RedirectedFolders	RedirectedFolders\$	Secondary site storage used for replication
G03SRV04	C:\DFS\RoamingProfiles	RoamingProfiles\$	Secondary site storage used for replication

Table 4: Printer Configurations

Printer, priority, and queue configurations

Printer Name	Server	Queue type	Permissions	Priority
k238 (HP LaserJet E40040)	G03SRV01VM	Standard	Administrators, Print Operators	99
k238-duplex	G03SRV01VM	Duplex	Domain Users	1

Table 5: WSUS Configurations

WSUS storage and update configurations

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Server	Storage Path	Synchronization Source	Updates Selected	Role
G03SRV07	F:\WSUS\	Microsoft Update	Windows 11, Windows Server 2025	GPO forces clients to retrieve updates from server